Assignment Question- Unit 1

Program: SY AIDS/AIML (Batch 2025-2029)	
Class : S.Y. B tech **Division & Batch : A&C**	**Course Name : DBMS** **Subject Code :U25DSPC302**

Q1.	**DBMS Fundamentals** Explain the architecture of a Database Management System (DBMS). Discuss the features of DBMS, describe the Three-Tier Architecture with a neat diagram, and compare DBMS with the traditional File System.
Q2.	**Entity-Relationship (ER) Modeling** Design an ER Diagram for a College Management System (or any suitable real-world application). Clearly identify the entities, attributes, relationships, cardinalities, and weak entities, and explain your design.
Q3.	**Extended ER (EER) Model** Explain the concepts of Generalization and Specialization in the Extended ER (EER) Model. Differentiate between them with suitable examples and illustrate each using neat diagrams.
Q4.	**Relational Algebra** Explain the Select (σ), Project (π), and Join ($\bowtie$) operations of Relational Algebra. Write the syntax and demonstrate each operation using a suitable example.
Q5.	**Database Design and Querying** Consider a Student-Course Management System. Design an ER model showing entities, attributes, cardinalities, and a weak entity. Extend the model using Generalization/Specialization, convert it into a relational schema, and write Relational Algebra expressions using Select, Project, and Join for any three queries based on your design.

Subject Teacher   Head
of Department